Sam Lokanc

EMAIL samlokanc5@gmail.com
PHONE 403-404-1063
WEBSITE https://samlokanc.github.io/

Relevant Skills

- **Python**
- **R**
- **SQL**
- **Data Visualization**
- **Geospatial Analysis**
- **Bash / Command Line**
- **Statistical Analysis**
- **Bayesian Modeling**
- **A/B Testing**
- **Hypothesis Testing**
- **Machine Learning**
- **Data Pipeline Engineering**

Experience

Bioinformatics Reasearch Associate **May 2024 - Mar 2026**
Arnie Charbonneau Cancer Institute, Calgary AB

- Led an independent bioinformatics research study examining the relationship between cancer and inborn errors of immunity (IEI)-related gene expression.
- Authored a peer-reviewed manuscript published in *Human Immunology.* https://doi.org/10.1016/j.humimm.2026.111671
- Presented findings a multiple academic conferences, earning **2nd place** in a competitive research presentation.
- Collaborated with faculty mentors to refine study design, analysis, and interpretation

Student Researcher **May 2023 - May 2024**
Arnie Charbonneau Cancer Institute, Calgary AB

- Selected for a competitive **O'brien Institute summer research grant** to conduct independent cancer immunotherapy research
- Designed **computational workflows** to support neoantigen-based vaccine development for ependymoma
- Conducted research under faculty supervision
- Presented and defended findings to academic audiences

Projects

Gene alterations in inborn errors of immunity and their presence in cancers **May 2024 - Mar 2026**
Research Project - Bioinformatics Research Associate Role

- Designed and implemented an RNA-seq analysis pipeline using **R and Python** to analyze differential gene expression across cancer and IEI-datasets
- Performed **data preprocessing, normalization, and statistical testing** in R and Python

Neoantigen Vaccine Design for Ependymoma **May 2023 - May 2024**
Undergraduate Thesis - University of Calgary

- Developed a **bash-based computational pipeline** on the University of Calgary high performance computing cluster to identify neoantigen vaccine candidates
- Integrated NetMHCpan and complementary **machine learning based prediction** tools to evaluate peptide-MHC binding affinity

Education

Master of Data Science **Sept 2025 - Present**
University of British Columbia, Vancouver, BC *Completion June 2026*

Bachelors of Health Science in Bioinformatics *(First Class Honours)* **Sept 2020 - May 2024**
University of Calgary, Calgary AB